Worksheet 8-1 A Average Value #1
AP Calculus

Name_____

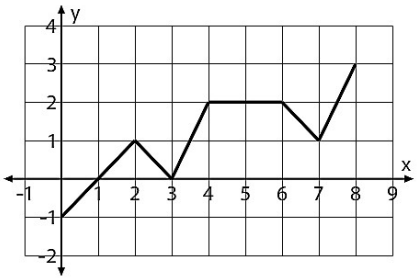
Directions: Find the average value of each function over the given interval

1.) $f(x) = 3x^2 + 8x$, $[-1, 2]$

2.) $f(x) = \sqrt{x}$, $[0, 4]$

3.) $g(t) = \dfrac{t}{\sqrt{3 + t^2}}$, $[1, 3]$

4.) Find the average value of f on $[0, 8]$



Directions: On the given interval, find the $x -$ value(s) where the function is equivalent to the average value on the interval.

5.) $f(x) = (x - 3)^2$, $[2, 5]$

6.) $f(x) = \frac{1}{x}$, $[1, 3]$

7.)

t (minutes)	0	4	8	12	16
$H(t)$ (°C)	65	68	73	80	90

The temperature in degrees Celsius (°C), of an oven being heated is modeled by an increasing, differentiable function H of time t, where t is measured in minutes. The table above gives the temperature as recorded every 4 minutes over a 16-minute period.

Estimate the average temperature of the oven over the interval $0 \le t \le 16$ using a left Riemann sum with four subintervals of equal length. Show the computations that lead to your answer.

Calculator-Active questions. Show the setup that leads to your answers.

8.) The temperature (in °F) t hours after 9AM is approximated by the function $T(t) = 50 + 14\sin\frac{\pi t}{12}$. Find the average temperature during the time period 9 AM to 9 PM.

9.) The depth of water in Mr. Brust's hot tub can be represented by the formula $h(t) = 2 - \cos t$, where t is the time in minutes since he begins pouring in water and $h(t)$ is measured in feet. What is the average depth of the water during the first three minutes.

10.) Traffic flow is defined as the rate at which cars pass through an intersection, measured in cars per minute. The traffic flow at a particular intersection is modeled by the function F defined by:

$$F(t) = 37 - 6\cos\left(\frac{t}{3}\right) \quad \text{for } 0 \le t \le 20$$

where $F(t)$ is measured in cars per minute and t is measured in minutes.

(a) What is the average traffic flow over the time interval $10 \le t \le 15$? Indicate units of measure.

(b) What is the average rate of change of the traffic flow over the time interval $10 \le t \le 15$? Indicate units of measure.